

PS9: Quarto Dashboard with Shiny

STAT 133

General Instructions

- Use the template file `ps9-template.qmd` to create your shiny dashboard. See bCourses `Files/problem-sets/ps9`.
- Rename the template file as `ps9-first-last.qmd`, where `first` and `last` are your first and last names (e.g. `ps9-gaston-sanchez.qmd`).
- Publish the dashboard to your **shinyapps.io** account.
- Submit both your qmd and shinyapps.io link to the corresponding assignment submission in bCourses.

About

In this assignment you will have to create a dashboard with "**shiny**". To be more precise, you will create a web-app for a future value calculator based on a fairly simple savings-and-investment scenario.

1 Savings-Investment Calculator

Say you are going to invest a lump-sum of \$1000 in some financial product that gives you an expected annual return r . Also, suppose you decide to make annual periodic contributions, e.g. \$720 at the end of each year. If the annual rate of return is $r = 8\%$, how much money would you expect to get at the end of a 10-year period? Or after 15 years? Or after 20 years?

To answer these questions, you can use the following formula to calculate the *Future Value* of your savings-investing scenario:

$$\text{FV} = \underbrace{P(1+r)^n}_{\text{lump sum future value}} + \underbrace{\text{PMT} \frac{(1+r)^n - 1}{r}}_{\text{ordinary annuity}}$$

where:

- FV = future value (amount accumulated)
- P = principal (how much you start with)
- PMT = periodic contribution made at the end of each period
- n = number of years
- r = annual interest rate (or annual rate of return)

1.1 Example: 10-year savings-investment scenario

Here's an example with the following **inputs** and outputs: **table** and **plot**.

Inputs

- initial investment of $P = \$1000$
- contributions of $\text{PMT} = \$720$ at the end of each year
- total number of years: $n = 10$
- annual rate of return: $r = 7.55\%$

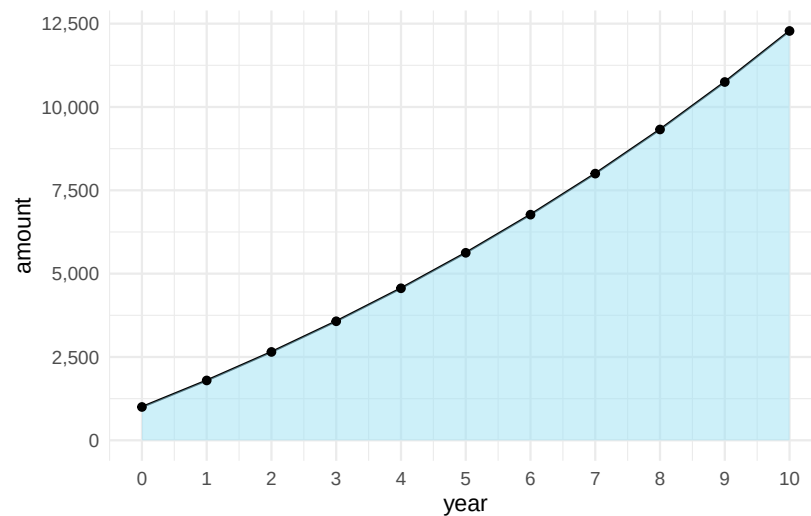
Output Table

Investment table for a 10-year period

	year	amount
initial investment →	0	1000.00
end-year 1 →	1	1795.50
	2	2651.06
	...	...
end-year 10 →	10	<i>etc</i>

Output Plot

Assuming that the data of the investment table is in a tibble (or data frame), we could graph a timeline like the following one:



2 Shiny Dashboard Layout

Your shiny dashboard should have a layout like the following diagram (see specifications below). BTW: we are providing a template dashboard—`ps9-template.qmd`—that comes with the necessary layout. Keep in mind that this file also contains dummy code that you need to replace with your own commands, and analysis ideas.

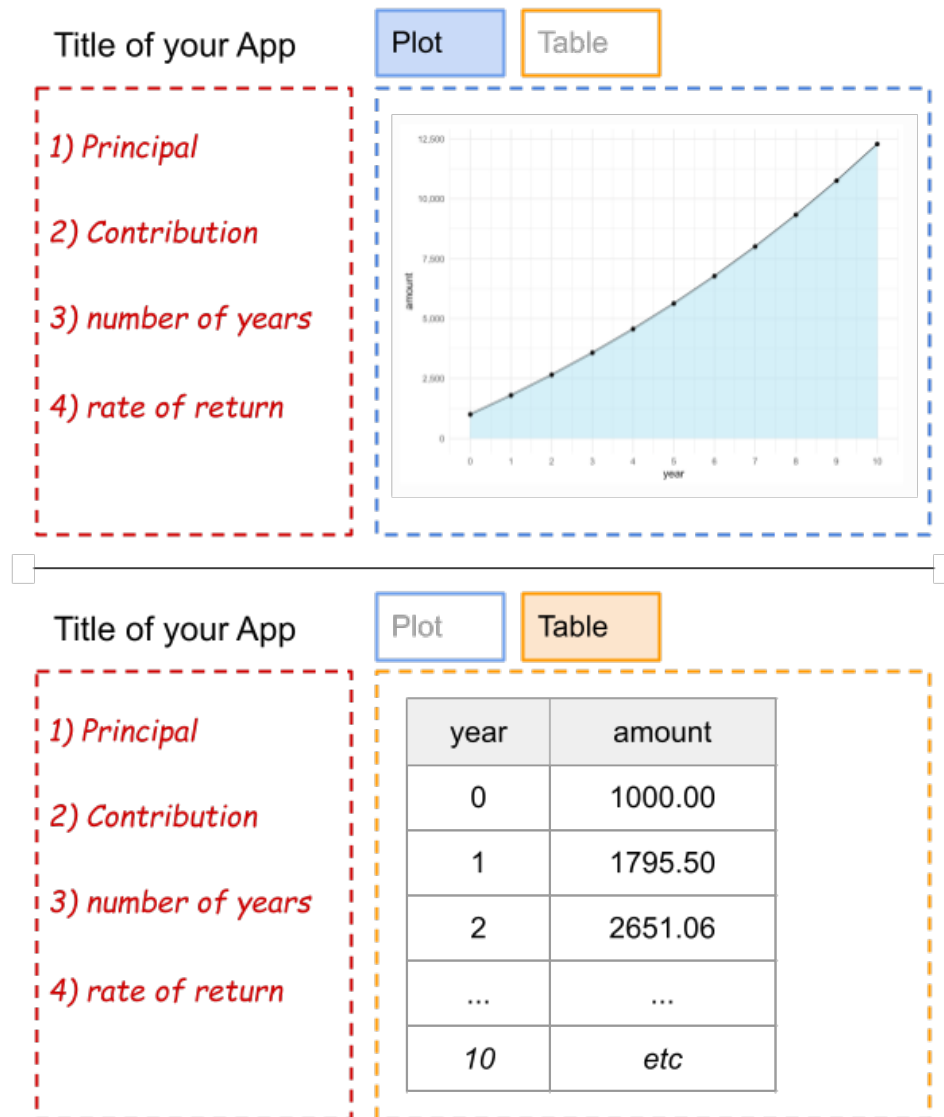


Figure 1: Layout of shiny dashboard with two tabs (Plot & Table)

As you can tell from the above diagram, the layout of the dashboard involves four distinctive sections—see file `ps9-template.qmd`:

- **Title**: main title for your app (give it a meaningful name).
- **Input Widgets**: widgets arranged in a sidebar panel.
- **Table**: a tab for an output table to display the balance of the investment in each year.
- **Plot**: a tab for an output graphic to display the timeline of the investment scenario.

2.1 Input widgets specifications

Your app should include widgets for the inputs listed below.

- 1) `numericInput()` or `sliderInput()` to specify the **initial amount** of the investment. Default value of 5000.
- 2) `numericInput()` or `sliderInput()` to specify the **annual contributions** of the investment. Default value of 1000.
- 3) `sliderInput()` for the **number of years**, ranging from 1 to 50. Default value of 20.
- 4) `numericInput()` or `sliderInput()` to specify the **rate of return**. Default value of 8% (i.e. 0.08).

2.2 Submission

Your submission to bCourses should include the following:

- qmd file
- Link to `shinyapps.io`